4.4 – Congruence and Transformations • Homework — Odd Answers

Name: _____

Name the single rigid motion that maps the preimage onto the image.

1. A(−3, 5) → A′(3, 5)

Answer: a reflection in the y-axis

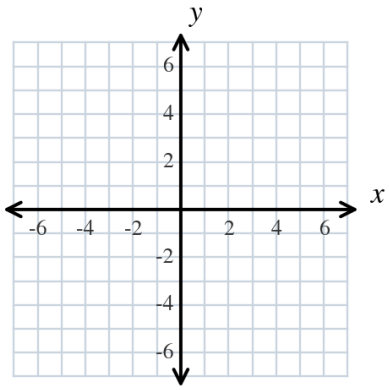
2. B(−3, 6) → B′(2, 4)

3. C(4, −7) → C′(−4, 7)

Answer: a 180° rotation about the origin

Find the final image after each composition.

4. D(2, −5) after a reflection in the x-axis, then the translation (x, y) → (x, y + 8).



5. △EFG with E(1, 1), F(4, 2), and G(3, −2) after a 90° rotation about the origin, then a reflection in the y-axis. Graph both.

Answer: after the rotation: E′(−1, 1), F′(−2, 4), G′(2, 3); after the reflection: E″(1, 1), F″(2, 4), G″(−2, 3)

Use the theorems about reflections in two lines.

6. Lines m: y = 2 and n: y = 5 are parallel. Reflect in m, then in n. Write ONE rule for the composition, and check it with (4, 1).

7. Reflect in the x-axis, then in the y-axis. What single transformation is this? Check it with (3, 1).

Answer: a 180° rotation about the origin; (3, 1) → (3, −1) → (−3, −1)

Answer each question.

8. A figure is reflected in two parallel lines 7 units apart. How far does the final image sit from the preimage, and in which direction?

9. Jo says a composition of three reflections cannot be a rigid motion because there are too many steps. Is Jo right? Explain.

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Answer: no — the Composition Theorem says any number of rigid motions compose to a rigid motion, so the figures are still congruent

10. △ABC ≅ △XYZ. Does that guarantee a congruence transformation maps △ABC onto △XYZ? Explain.